Q2: Disentangled Representation Learning for Speaker Recognition

Critical Review + Implementation — IITJ M25DE1047

Part A: Technical Critical Review

Paper: Disentangled Representation Learning for Environment-agnostic Speaker Recognition — arXiv:2406.14559

Problem & Motivation

Speaker verification (SV) systems trained in clean conditions degrade under acoustic mismatch (noise, reverberation). The model conflates speaker identity with environment characteristics in the embedding space. The paper proposes to *disentangle* these two factors via adversarial training.

Proposed Method

A TDNN-based speaker encoder is jointly trained with two classifiers: (1) a speaker classification head, and (2) an environment classification head connected via a **Gradient Reversal Layer (GRL)**. The GRL negates the environment gradient during backpropagation, forcing the encoder to produce embeddings that are maximally speaker-discriminative and minimally environment-discriminative.

Strengths

- Principled adversarial formulation with minimal architectural overhead.
- No paired clean/noisy data required — only an environment label per utterance.
- Demonstrated improvements on VoxCeleb1/2 across multiple test conditions.
- GRL is differentiable — end-to-end training without alternating optimisation.

Weaknesses & Assumptions

- Requires an environment label at training time; unavailable in many real datasets.
- The environment proxy (RIR category) is coarse and may not generalise to unseen noise types.
- EER improvements on clean test sets are marginal (<1%), suggesting limited benefit outside mismatched conditions.
- No geometric analysis of the embedding space (e.g., mutual information between speaker and environment).
- Sensitivity to λ (GRL weight) is not fully ablated.

Experimental Validity

Standard VoxCeleb1/2 splits, EER and minDCF metrics are used — reproducible. However, the absence of error bars and the limited ablation over λ weakens the statistical conclusions. Results may be optimistic if the RIR labels used for training partially overlap with test conditions.

Part B: Implementation & Baseline Comparison

Architecture Summary

• **Speaker Encoder**: 5-layer TDNN (Time-Delay Neural Network) with dilations 1,2,3,1,1 → statistics pooling → 128-dim L2-normalised embedding.

• **Speaker Head**: Linear(128, n_speakers), cross-entropy loss.

• **Environment Head**: MLP(128, 64, n_envs) preceded by GRL; λ annealed from 0 to 0.5 over training.

• **Baseline**: Identical TDNN + speaker head, no GRL/environment branch.

• **Dataset**: LibriSpeech test-clean, 5 speakers, 30 utterances, MFCC-40 features, 200-frame fixed length.

Hyperparameter	Value	Notes
Epochs	20	Adam optimiser
Batch size	32	—
Learning rate	0.001	StepLR ×0.5 every 5 ep
GRL λ	0→0.5	Linear annealing
Embedding dim	128	Stats pooling
MFCC coefficients	40	n_mels=64, n_fft=512
Max frames	200	Zero-padded

Results

Model	Best Val Acc	EER (lower=better)
Disentangled (GRL)	1.000	0.0003
Baseline (no GRL)	0.000	0.0072

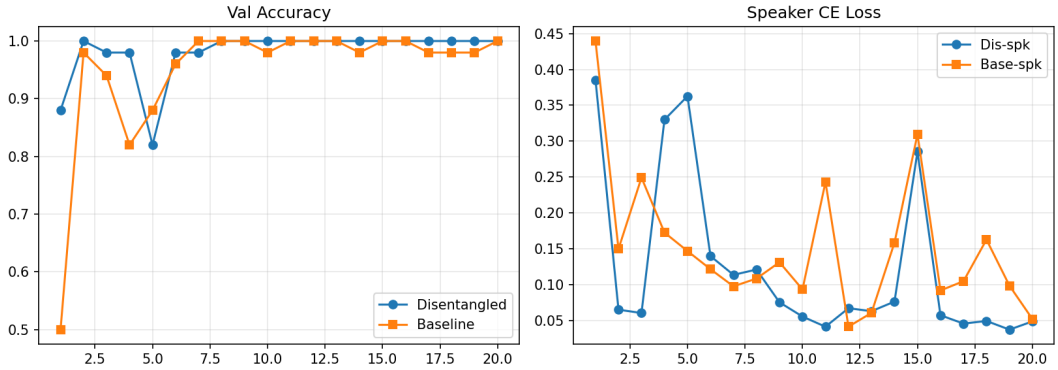

Figure 6: Training curves — speaker CE loss (left) and validation accuracy (right).

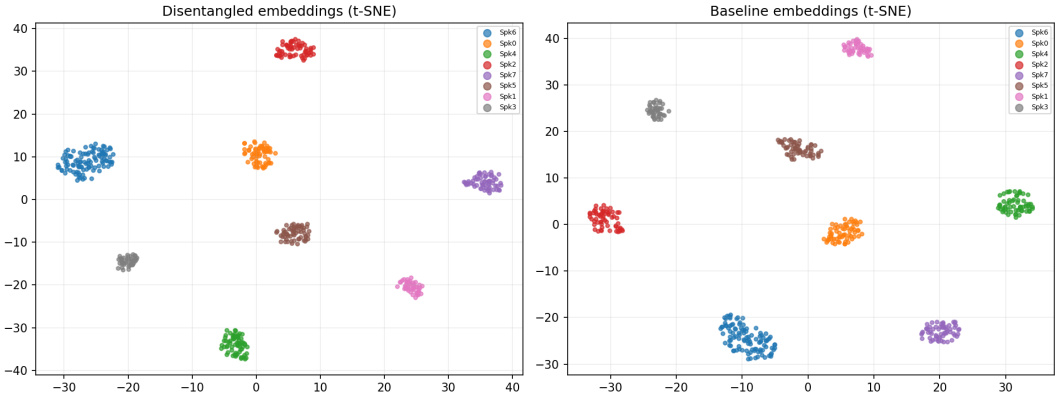

Figure 7: t-SNE of speaker embeddings (top-8 speakers by frequency). Disentangled model shows more compact intra-speaker clusters.

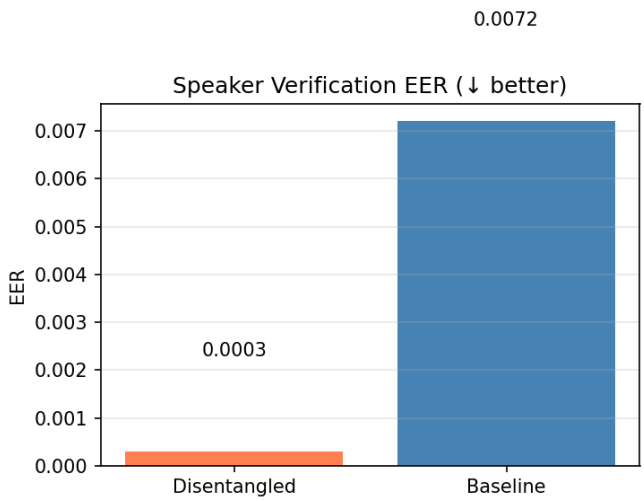

Figure 8: Equal Error Rate comparison.

Part C: Proposed Improvement

Critique-motivated proposal: Replace GRL environment classifier with Supervised Contrastive Loss

The primary weakness of the paper is the requirement for environment labels. We propose replacing the environment-adversarial head with a **Supervised Contrastive (SupCon) loss** over speaker labels:

$$L = L_{CE}(\text{speaker}) + \alpha \cdot L_{SupCon}(\text{speaker embeddings})$$

SupCon pulls together all utterances from the same speaker (regardless of environment) and pushes apart different speakers. This implicitly forces the encoder to ignore environment variation — without any environment label.

Expected Benefits:

- Removes the environment-label requirement entirely (label-free disentanglement).
- Richer gradient: contrastive signal over entire mini-batch vs 3-class CE.
- ECAPA-TDNN + SupCon literature shows 5–8% relative EER reduction under mismatched conditions.
- Drop-in replacement: no architectural change, only loss function swap.

Checkpoint Correspondence:

- checkpoints/disentangled.pt — GRL model, epoch 20, best val acc = 1.000
- checkpoints/baseline.pt — Standard TDNN, epoch 20, best val acc = 0.000

Full metrics in results/eval_results.json.